SpinShelf

A motorized rotating cabinet that brings stuff from the back of the shelf to you

Team 15, Everyday Reach, DES INV 15, Fall 2026

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everyday-reach.vercel.app

The problem: reaching around the house causes falls

1 in 4

adults 65+ fall every year (CDC)

3.85M

older adults went to the ER for a fall in 2023 (NSC)

28%

of adults 65+ live alone (Pew, 2025)

Who we're designing for

Margaret, 72

- Retired teacher, lives alone in Berkeley
- Low cabinets and high shelves are getting hard
- Her grabber can't lift heavier stuff, so she waits for her daughter
- Uses a smart speaker, but wants physical buttons as a backup

"I just want to be able to reach the things I need without worrying about hurting myself."

How might we statements

Main question

How might we help older adults safely get everyday household objects without a lot of bending or reaching?

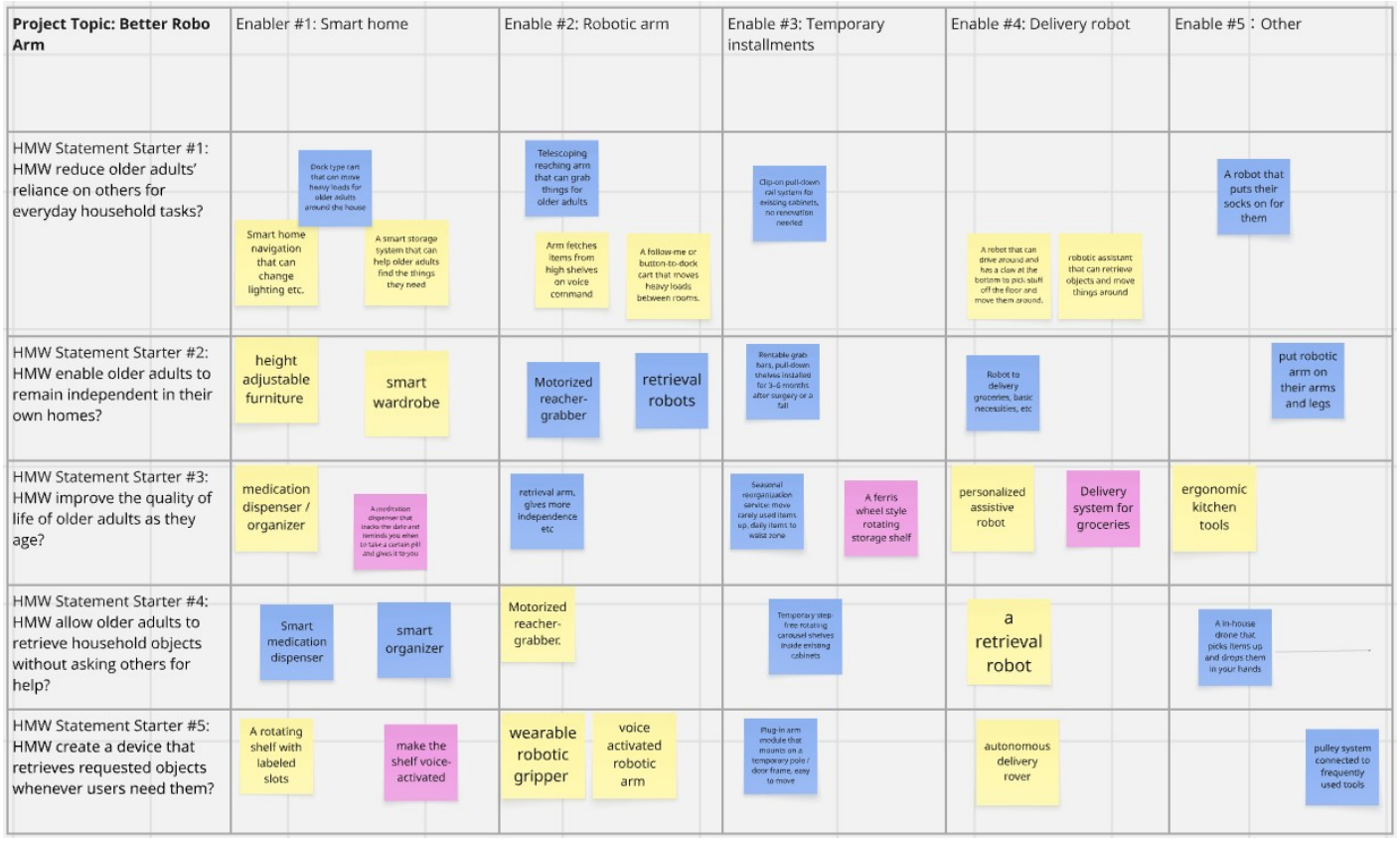
HMW help older adults living alone reach everyday objects without risking a fall?

HMW make everyday items easy to reach wherever they're stored?

HMW help older adults stay independent without relying on others for small tasks?

HMW make getting things easier without major home renovations?

How we got to the concept



Project Topic: Better Robo Arm	Enabler #1: Smart home	Enable #2: Robotic arm	Enable #3: Temporary installments	Enable #4: Delivery robot	Enable #5 : Other
HMW Statement Starter #1: HMW reduce older adults' reliance on others for everyday household tasks?	Dock type cart that can move heavy loads for older adults around the house; Smart home navigation that can change lighting etc.; A smart storage system that can help older adults find the things they need	Telescoping reaching arm that can grab things for older adults; Arm fetches items from high shelves on voice command; A follow-me or button-to-dock cart that moves heavy loads between rooms.	Clip-on pull-down rail system for existing cabinets, no renovation needed	A robot that can drive around and has a claw at the bottom to pick stuff off the floor and move them around.; robotic assistant that can retrieve objects and move things around	A robot that puts their socks on for them
HMW Statement Starter #2: HMW enable older adults to remain independent in their own homes?	height adjustable furniture; smart wardrobe	Motorized reacher-grabber; retrieval robots	Rentable grab bars, pull-down shelves installed for 3–6 months after surgery or a fall	Robot to delivery groceries, basic necessities, etc	put robotic arm on their arms and legs
HMW Statement Starter #3: HMW improve the quality of life of older adults as they age?	medication dispenser / organizer; A medication dispenser that tracks the date and reminds you when to take a certain pill and gives it to you	retrieval arm, gives more independence etc	Seasonal reorganization service: move rarely used items up, daily items to waist zone; A ferris wheel style rotating storage shelf	personalized assistive robot; Delivery system for groceries	ergonomic kitchen tools
HMW Statement Starter #4: HMW allow older adults to retrieve household objects without asking others for help?	Smart medication dispenser; smart organizer	Motorized reacher-grabber.	Temporary step free rotating carousel shelves inside existing cabinets	a retrieval robot	A in-house drone that picks items up and drops them in your hands
HMW Statement Starter #5: HMW create a device that retrieves requested objects whenever users need them?	A rotating shelf with labeled slots; make the shelf voice-activated	wearable robotic gripper; voice activated robotic arm	Plug-in arm module that mounts on a temporary pole / door frame, easy to move	autonomous delivery rover	pulley system connected to frequently used tools

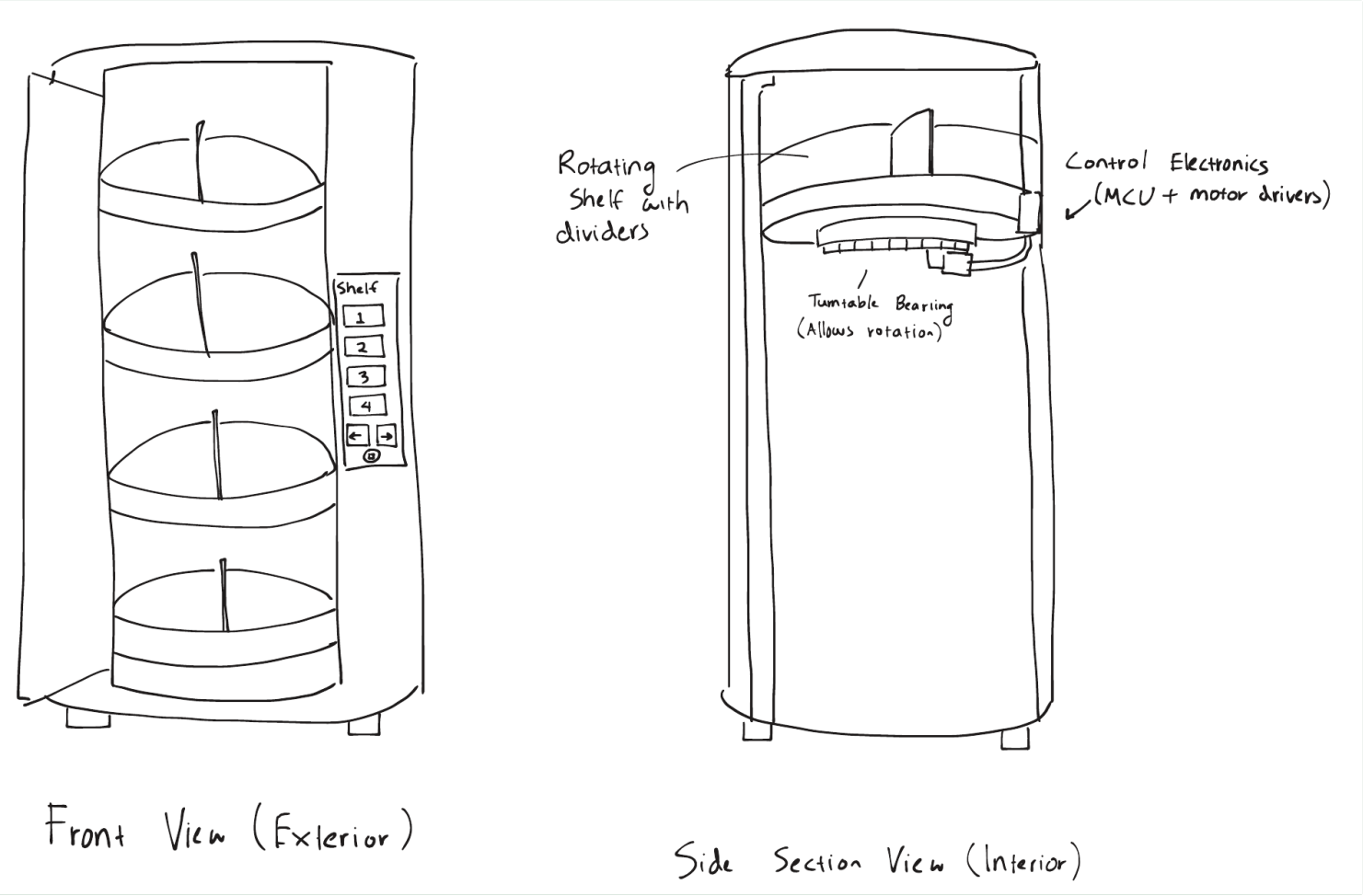
Our creative matrix (HMW statements vs. enablers)

We started out wanting a robotic arm.

- Arms are expensive, hard to build well, and kind of intimidating in someone's kitchen
- Gripping random objects reliably is a hard problem on its own
- The rotating shelf ideas on the board solved the same problem with way fewer moving parts

So we switched to a rotating cabinet: SpinShelf.

SpinShelf: our concept sketches



Front view (left) and side section view (right)

How it works

- Tall round cabinet with a door and 4 shelves stacked inside
- Each shelf sits on a turntable bearing, with dividers splitting it into sections
- Side panel: pick shelf 1 to 4, then use the arrows to spin it
- A motor turns the shelf so stuff in the back comes to the front
- Control electronics (microcontroller + motor drivers) live in the side wall

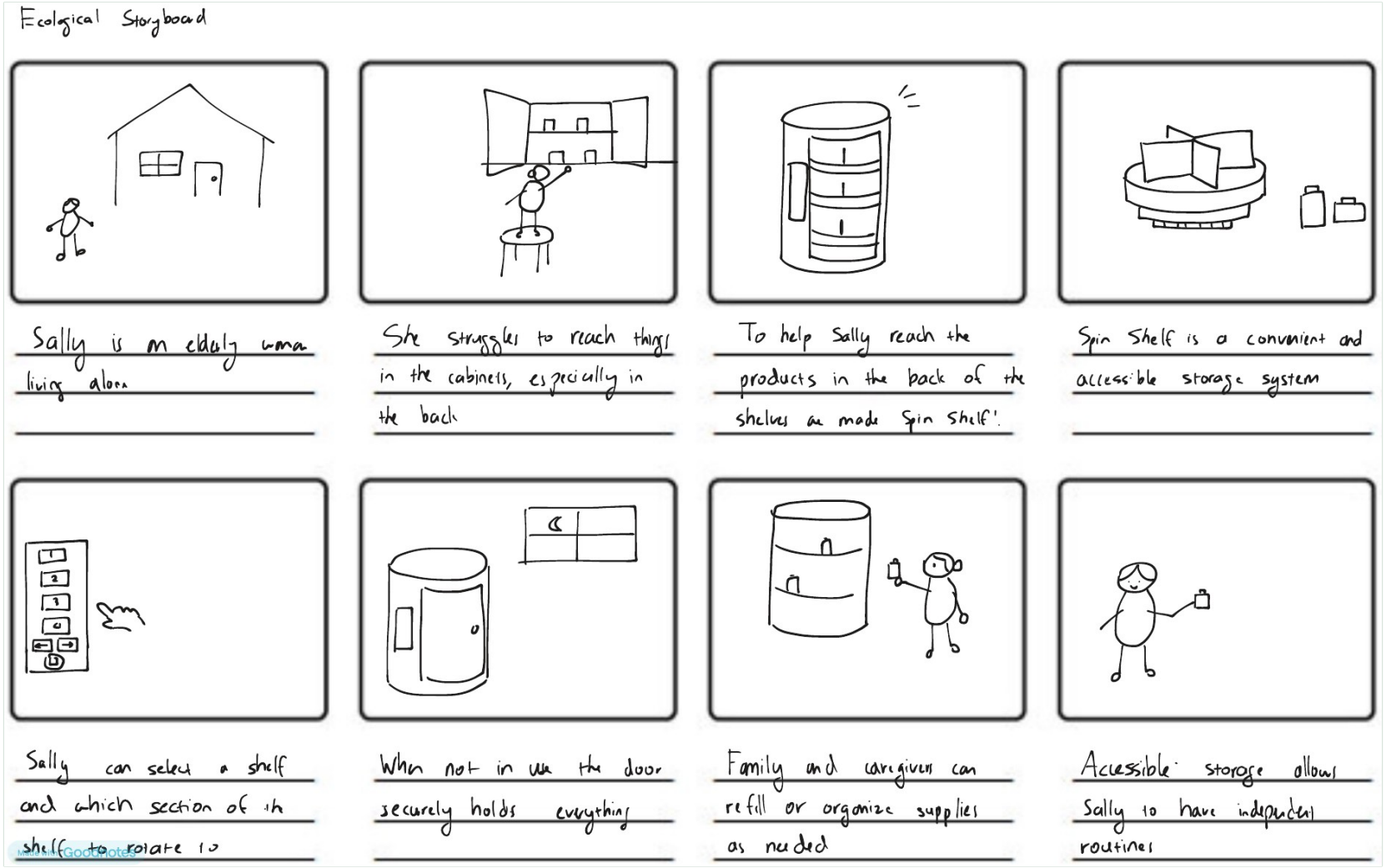
Interaction storyboard



How Sally uses it

- Sally just started living alone and can't reach the back of her cabinets
- She gets SpinShelf installed
- She picks a shelf on the side panel and it spins things from the back to the front

Ecological storyboard



Where it fits in her life

- Pick a shelf and which section to rotate to
- When it's not in use, the door keeps everything closed up
- Family or caregivers can restock it when they visit
- Sally gets to keep her own routines

What the prototype needs to show and teach us

What it has to show users

- Pick a shelf, and the back of it spins around to you
- It's basically a lazy Susan that turns itself, nothing fancy
- It moves slowly and won't pinch or surprise you
- If it breaks you can still turn it by hand

What we need to learn

Our biggest worry: will older adults actually trust something that moves on its own? A lot of them didn't grow up with this kind of tech.

- Does the speed and noise feel safe or sketchy?
- Would they use it over counters and grabbers?
- Does the shelf + arrow panel make sense right away?
- What would they store in it, and how heavy?
- What would they want to happen if it jams?

How we'll prototype it, and what's next

Works-like prototype

- Start with one or two shelves: lazy Susans with dividers inside a cardboard or foam core cylinder
- Arduino + motor, with a mini version of the side panel (shelf buttons + arrows)
- If the motor isn't ready, one of us turns it by hand behind the box when a button gets pressed (Wizard of Oz), so we can still test
- Big, clearly labeled buttons so we can test if the panel makes sense

Testing

We'll have older adults do simple tasks like "get the cinnamon from the back of shelf 2," watch where they hesitate, then talk to them about whether they'd trust it at home.

Next steps

Nov 9 to 15

Lock in the design, number of shelves, and panel layout

Nov 16 to 22

Build the low-fi prototype

Nov 23 to 29

Test with older adults and iterate

Nov 30 to Dec 6

Final prototype, update the site, prep final presentation

Who's doing what

MP

Minseo Park

Hardware

- Builds the cabinet body and rotating shelves
- Figures out the physical mechanism
- Looks-like details (size, labels, buttons)

L

Leo

Hardware

- Wires the Arduino, motor, and side panel
- Makes sure it turns smoothly and stops in the right spot
- Manual turn backup

AK

Aadi Kulkarni

Software + coordination

- Writes the panel and motor control code
- Sets up user testing sessions
- Runs the timeline, website, and presentation

Everyone helps with interviews and testing.